

(a) Fixed model and inputs

Training participants
signals and labels

Frozen classifier f
LR / MiniRocket / IT

Monitoring batch X^* (M unlabeled inputs)
all candidate-label scores
 $s(x,y)$ available; labels hidden

prediction sets $C(x; \bar{U}_T)$

monitoring scores

(c) Sequential querying and prediction-set stopping

Fixed pool P : N windows
true labels hidden
uniform random query order

Query next 20 labels
reveal true-label scores
sort observed prefix

Update rank bracket
running intersection:
 $L_t = \max L_v, \bar{U}_t = \min U_v$

$D_t \leq \tau$?

Return $C(x; \bar{U}_T)$
stop label queries

$$D_t = (1/M) \sum_{x \in X^*} (|C(x; \bar{U}_t)| - |C(x; L_t)|)$$

No: next query batch

(b) Label-free rank calibration — offline, before any query

Inputs: N , $k = \lceil (N+1)(1-\alpha) \rceil$, δ , checkpoints T
Joint (J_t, R_t) rank-path recursion — Eqs. (7)–(8)
Outputs calibrated boundaries (ℓ_t, u_t) at every checkpoint

Guarantee (prob $\geq 1-\delta$ over query order)

$$C(x; q_{ref}) \subseteq C(x; \bar{U}_T) \quad \text{for every } x$$

$$(1/M) \sum_x (|C(x; \bar{U}_T)| - |C(x; q_{ref})|) \leq \tau$$